

1.Installation Guide for MicroPython Kernel and Jupyter Notebook (Windows OS)

Prerequisites

Before starting, ensure you have the following:

- **Operating System:** Windows 10 or later
- **Internet Browser:** Google Chrome (recommended)
- **Stable Internet Connection**
- **Basic knowledge of Windows OS**

Step 1: Install Python

1. Open **Browser** and go to the [Python downloads page](https://www.python.org/downloads/)-.<https://www.python.org/downloads/>
2. Click on **Download Python 3.13** (or the latest available version).
3. Choose a directory (e.g., **Downloads**) and click **OK** to start downloading.
4. Navigate to the **Downloads** folder and double-click the **Python executable file**.
5. In the installation window:
 - Check **"Add Python.exe to PATH"**.
 - Click **Install Now**.
6. Wait for the installation to complete and look for the success message.

Verify Python Installation

1. Open **Command Prompt** (Press **Win + S**, type **cmd**, and press **Enter**).
2. Type the following command and press **Enter**:
`python --version`
3. The installed Python version (e.g., **Python 3.13**) should be displayed.

Step 2: Install Jupyter Notebook

1. Update **pip** (Python's package manager) by running the command:
`python -m pip install --upgrade pip`
2. Install Jupyter Notebook using:
`python -m pip install jupyter`
3. Wait for the installation to finish. You should see a success message.

Step 3: Install the MicroPython Kernel

1. Open **Command Prompt** and type:
`pip3 install jupyter_micropython_kernel`
2. Register the MicroPython kernel with Jupyter by running:
`python -m jupyter_micropython_kernel.install`
3. Verify the installation by checking the available kernels:
`jupyter kernelspec list`
You should see the MicroPython **Kernel** in the list.

Step 4: Upload MicroPython Firmware to ESP32

Download the Firmware

1. Open Google Chrome and visit the [MicroPython Firmware download page](https://micropython.org/download/ESP32_GENERIC/).-https://micropython.org/download/ESP32_GENERIC/
2. Click on the latest **.bin file** to download it.

Flash the Firmware onto ESP32

- Connect the **ESP32 microcontroller** to your computer using a **USB cable**.
- Open a new tab in Chrome and go to [Adafruit WebSerial ESPTool](https://adafruit.github.io/Adafruit_WebSerial_ESPTool/).-https://adafruit.github.io/Adafruit_WebSerial_ESPTool/
- Click the **Connect** button in the top-right corner.
- Select the USB port where ESP32 is connected and click **Connect**.
- Press and hold the **BOOT button** on the ESP32 board.
- In the tool interface:
 - Enter **1000** in the first offset text box.
 - Click **Choose File** and select the **ESP32 firmware .bin file** from your download directory.
 - Click **Program** to upload the firmware.

Wait for the upload to complete. A message will appear: *"To run the new firmware, please reset your device."*

2.Installation Guide for MicroPython Kernel on Jupyter Notebook (Linux OS)

Prerequisites

Before starting, ensure you have the following:

- **Operating System:** Ubuntu Linux OS (version 22.04 or later)
- **Internet Browser:** Google Chrome (recommended)
- **Stable Internet Connection**
- **ESP32 Microcontroller**
- **Basic knowledge of Ubuntu Linux OS**

Step 1: Install Python

1. Open a **Terminal window** (**Ctrl + Alt + T**).
- Install Python by typing the following command:
`sudo apt install python3`
- 2.
3. Press **Enter** and enter your **system password** when prompted.
4. The terminal will display a list of packages to install.
5. Press **Y** and then **Enter** to confirm the installation.
6. Wait for the installation to complete.

Verify Python Installation

- In the terminal, type the following command and press **Enter**:
`python3 --version`
- 1.
- 2. The installed Python version should be displayed.

Step 2: Install Jupyter Notebook

- Update **pip** (Python's package manager) by running the command:
`python3 -m pip install --upgrade pip`
- 1.
- Install Jupyter Notebook using:
`pip3 install notebook`
- 2.
- 3. Wait for the installation to finish. You should see a success message.

Step 3: Install the MicroPython Kernel

- Open **Terminal** and type:
`pip3 install jupyter_micropython_kernel`
- 1.
- Register the MicroPython kernel with Jupyter by running:
`python3 -m jupyter_micropython_kernel.install`
- 2.
- Verify the installation by checking the available kernels:
`jupyter kernelspec list`
- 3. You should see the MicroPython **Kernel** in the list.

Step 4: Upload MicroPython Firmware to ESP32

Download the Firmware

1. Open Google Chrome and visit the [MicroPython Firmware download page](#).
2. Click on the latest **.bin file** to download it.

Flash the Firmware onto ESP32

1. Connect the **ESP32 microcontroller** to your computer using a **USB cable**.
2. Open a new tab in Chrome and go to [Adafruit WebSerial ESPTool](#).
3. Click the **Connect** button in the top-right corner.
4. Select the USB port where ESP32 is connected and click **Connect**.
5. Press and hold the **BOOT button** on the ESP32 board.
6. In the tool interface:
 - Enter **1000** in the first offset text box.
 - Click **Choose File** and select the **ESP32 firmware .bin file** from your download directory.
 - Click **Program** to upload the firmware.
7. Wait for the upload to complete. A message will appear: "To run the new firmware, please reset your device."